# Advanced Machine Learning

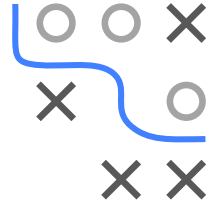
# Multi-Target Prediction: Introduction

| | | Tennis | Football | Biking | Movies | TV | Belgium |
|---|---|---|---|---|---|---|---|
| 01101 | Text1 | 0 | 1 | 0 | 0 | 1 | 1 |
| 00111 | Text2 | 1 | 0 | 0 | 0 | 0 | 1 |
| 01110 | Text3 | 0 | 0 | 0 | 1 | 1 | 0 |
| 10001 | Text4 | 0 | 0 | 1 | 0 | 1 | 0 |
| 01011 | Text5 | 1 | 0 | 0 | 1 | 0 | 0 |
| 11110 | Text6 | ? | ? | ? | ? | ? | ? |

## Learning goals

- Understand the practical relevance of multi-target prediction problems
- Know relevant special cases of multi-target prediction
- Understand the difference between inductive and transductive learning problems

# MULTI-TARGET PREDICTION: MOTIVATION

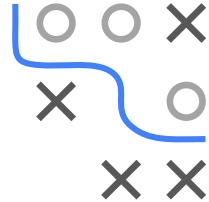


- Conventional supervised learning: Label space $\mathcal{Y}$ is 1-D.
- Multi-target prediction (MTP): multiple targets of mixed types (binary, nominal, ordinal, real-valued).
- Learn one model per target independently? $\rightsquigarrow$ Targets can be *statistically dependent*.
- Multi-label Emotions Dataset: 4 emotions of a music piece. Multiple emotions may be attributed to a single piece. Mutual information of the labels are:

| | **Calm** | **Quiet** | **Sad** | **Angry** |
|---|---|---|---|---|
| **Calm** | 1.000 | 0.073 | 0.018 | **0.290** |
| **Quiet** | 0.073 | 1.000 | **0.241** | **0.164** |
| **Sad** | 0.018 | **0.241** | 1.000 | 0.067 |
| **Angry** | **0.290** | **0.164** | 0.067 | 1.000 |

- It might be better to tackle targets *simultaneously*.

# MULTI-TARGET PREDICTION: CHARACTERISTICS



Characterized by instances $\mathbf{x} \in \mathcal{X}$ and targets $m \in \{1, 2, \ldots, l\}$ with following properties:

- A training set $\mathcal{D} = \{(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})\}_{i=1}^{n}$, where $\mathbf{y}^{(i)} = (y_1^{(i)}, \ldots, y_l^{(i)})$, with $y_m^{(i)} \in \mathcal{Y}_m$ is label for target $m$.
- $n$ instances and $l$ targets $\rightsquigarrow$ Labels $y_m^{(i)}$ can be arranged in an $n \times l$ matrix $\boldsymbol{Y}$. Note $\boldsymbol{Y}$ may have missing values.
- Target spaces $\mathcal{Y}_m$ can be nominal, ordinal or real-valued.
- Goal: predict scores for any pair $(\mathbf{x}, m) \in \mathcal{X} \times \{1, 2, \ldots, l\}$.

In conventional MTP setting: no available side information for targets.

# MULTIVARIATE REGRESSION

Target space $\mathcal{Y}_m = \mathbb{R}\ \forall m \in \{1, 2, \ldots, l\}$ .

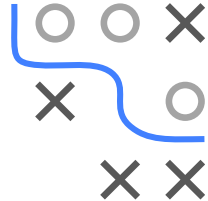
| | Mol1 | Mol2 | Mol3 | Mol4 | Mol5 | Mol6 |
|---|---|---|---|---|---|---|
| 01101 | 1,3 | 0,2 | 1,4 | 1,7 | 3,5 | 1,3 |
| 00111 | 2 | 1,7 | 1,5 | 7,5 | 8,2 | 7,6 |
| 01110 | 0,2 | 0 | 0,3 | 0,4 | 1,2 | 2,2 |
| 10001 | 3,1 | 1,1 | 1,3 | 1,1 | 1,7 | 5,2 |
| 01011 | 4,7 | 2,1 | 2,5 | 1,5 | 2,3 | 8,5 |
| 11110 | ? | ? | ? | ? | ? | ? |

Waegeman et al. (2019), Multi-target prediction: A unifying view on problems and methods (URL).

Example: Predict binding strength between proteins (rows) and molecules (columns).

# MULTI-LABEL CLASSIFICATION

Target space $\mathcal{Y}_m = \{0, 1\}\ \forall m \in \{1, 2, \ldots, l\}$



| | | Tennis | Football | Biking | Movies | TV | Belgium |
|---|---|---|---|---|---|---|---|
| 01101 | Text1 | 0 | 1 | 0 | 0 | 1 | 1 |
| 00111 | Text2 | 1 | 0 | 0 | 0 | 0 | 1 |
| 01110 | Text3 | 0 | 0 | 0 | 1 | 1 | 0 |
| 10001 | Text4 | 0 | 0 | 1 | 0 | 1 | 0 |
| 01011 | Text5 | 1 | 0 | 0 | 1 | 0 | 0 |
| 11110 | Text6 | ? | ? | ? | ? | ? | ? |

Waegeman et al. (2019), Multi-target prediction: A unifying view on problems and methods (URL).

Example: Assign documents (rows) to category tags (columns).

# LABEL RANKING

In *label ranking*, each instance is associated with a ranking of targets. $\mathcal{Y}_m = \{1, \ldots, l\}\ \forall m$, and labels (i.e., ranks) $y_m^{(i)} \neq y_k^{(i)} \forall m \neq k$.

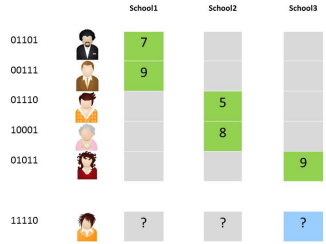
| | | Tennis | Football | Biking | Skating | Running | Walking |
|---|---|---|---|---|---|---|---|
| 01101 | User 1 | 2 | 1 | 4 | 3 | 5 | 6 |
| 00111 | User 2 | 1 | 4 | 3 | 5 | 6 | 2 |
| 01110 | User 3 | 4 | 5 | 1 | 2 | 3 | 6 |
| 10001 | User 4 | 4 | 3 | 2 | 6 | 1 | 5 |
| 01011 | User t5 | 1 | 3 | 5 | 2 | 6 | 4 |
| 11110 | User 6 | ? | ? | ? | ? | ? | ? |

Waegeman et al. (2019), Multi-target prediction: A unifying view on problems and methods (URL).

Example: Predict for users (rows) their preferences over specific activities (columns).

# MULTI-TASK LEARNING

- **Not all targets are relevent for all instances**. E.g., a student may only attend one school, other labels are **irrelavent**.
- Label space is homogenous across columns of $\boldsymbol{Y}$, e.g., $\mathcal{Y}_m = \{0, 1\}$ or $\mathcal{Y}_m = \mathbb{R}$ for all $m$.

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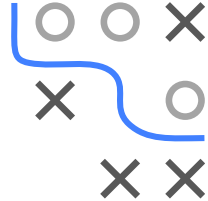
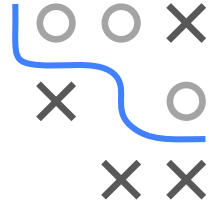


Example: Predict for students (rows) the final grades for a specific high-school course (columns).

# REMARKS

- It is also possible when the $m$-th task is multiclass classification. That is, $f(\mathbf{x})_m \in \mathbb{R}^{g_m}$ is the probability predictions for $g_m$ classes.
  $\rightsquigarrow$ The techniques for multi-target learning are also applicable under this setting, notation becomes cumbersome.
- Target space can be inhomogeneous, e.g. $\mathcal{Y}_m = \{0, 1\}$ and $\mathcal{Y}_k = \mathbb{R}$.
  $\rightsquigarrow$ A mixture of multi-label classification and multivariate regression.



# SIDE INFORMATION ON TARGETS

- Sometimes, additional side information about targets is available.

- Extra representation for target molecules in drug design (*structured representation*).

| | Mol1 | Mol2 | Mol3 | Mol4 | Mol5 | Mol6 |
|---|---|---|---|---|---|---|
| 01101 | 1,3 | 0,2 | 1,4 | 1,7 | 3,5 | 1,3 |
| 00111 | 2 | 1,7 | 1,5 | 7,5 | 8,2 | 7,6 |
| 01110 | 0,2 | 0 | 0,3 | 0,4 | 1,2 | 2,2 |
| 10001 | 3,1 | 1,1 | 1,3 | 1,1 | 1,7 | 5,2 |
| 01011 | 4,7 | 2,1 | 2,5 | 1,5 | 2,3 | 8,5 |
| 11110 | ? | ? | ? | ? | ? | ? |

- Taxonomy on document categories (*hierarchy*).

| | | | | | | | |
|---|---|---|---|---|---|---|---|
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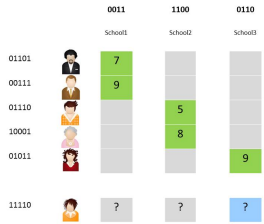
| 00111 | Text2 | 0 | 0 | 1 | 0 | 1 | 1 |
| 01110 | Text3 | 0 | 0 | 0 | 1 | 1 | 0 |
| 10001 | Text4 | 0 | 0 | 1 | 0 | 1 | 0 |
| 01011 | Text5 | 1 | 0 | 0 | 1 | 0 | 0 |
| 11110 | Text6 | ? | ? | ? | ? | ? | ? |

# SIDE INFORMATION ON TARGETS / 2

- Information about schools (geographical location, school reputation) in student mark forecasting (*feature representation*).




Waegeman et al. (2019), Multi-target prediction: A unifying view on problems and methods (URL).

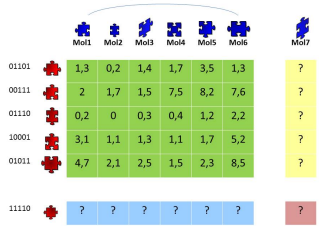
- Such problems are referred to as dyadic or link prediction.
- Labels $y_m^{(i)}$ can be arranged in a matrix $\boldsymbol{Y}$, which is often sparse.
- Thus, *dyadic prediction* can be seen as *multi-target prediction with target features*.

# INDUCTIVE VS. TRANSDUCTIVE LEARNING

- In previous problems,
  1. predictions need to be generated for novel instances,
  2. targets are known beforehand and observed during training.
- These problems are *inductive* w.r.t. instances and *transductive* w.r.t. targets.

- Side information is important for generalizing to novel targets.
  - a novel target molecule in the drug design,
  - a novel tag in the document annotation,

$g(.,.)$ : target similarity



| | Mol1 | Mol2 | Mol3 | Mol4 | Mol5 | Mol6 | Mol7 |
|---|---|---|---|---|---|---|---|
| 01101 | 1,3 | 0,2 | 1,4 | 1,7 | 3,5 | 1,3 | ? |
| 00111 | 2 | 1,7 | 1,5 | 7,5 | 8,2 | 7,6 | ? |
| 01110 | 0,2 | 0 | 0,3 | 0,4 | 1,2 | 2,2 | ? |
| 10001 | 3,1 | 1,1 | 1,3 | 1,1 | 1,7 | 5,2 | ? |
| 01011 | 4,7 | 2,1 | 2,5 | 1,5 | 2,3 | 8,5 | ? |
| 11110 | ? | ? | ? | ? | ? | ? | ? |

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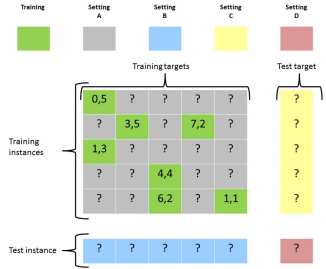
# SUBDIVISION OF DIFFERENT LEARNING SETTINGS

- Setting A — transductive w.r.t. targets and instances. Goal: predict missing values of score matrix (*matrix completion*).
- Setting B — transductive w.r.t. targets and inductive w.r.t. instances (*classical supervised learning*).
- Setting C — inductive w.r.t. targets and transductive w.r.t. instances. $\rightsquigarrow$ Some targets are unobserved during training but may appear at prediction time.
- Setting D — inductive w.r.t. both targets and instances (*zero-shot learning*).

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